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NMAM INSTITUTE OF TECHNOLOGY, NITTE
Off-Campus Centre of Nitte (Deemed to be University)
Seventh Semester B.Tech (CBCS) Degree Examinations

December 2025
AD2313-1 – UI AND UX DESIGN
(For AD)

Duration: 3 Hours

Max. Marks: 100

Note:

Part – A: Multiple Choice Questions: Answer all **Twenty** questions in the **OMR Sheet** provided. Each question carries equal marks.

Part – B: Descriptive Answer Questions: Answer **Five full** questions choosing **Two full** questions from **Unit – I & Unit – II** each and **One full** question from **Unit – III**.

PART - A: MULTIPLE CHOICE QUESTIONS

20 Marks

1. Which of the following best describes UI design?
A) Focuses on how a product feels B) Involves the functional architecture of a system
C) Focuses on the visual layout and interactive elements D) Deals with understanding user emotions
2. UI design is primarily concerned with:
A) Usability testing B) Typography, colors, and layouts
C) Long-term emotional attachment D) Strategic business goals
3. Which of the following pairs matches correctly?
A) UI – Appearance, UX – Functionality Experience B) UI – Research, UX – Visual Styling
C) UI – Mental models, UX – Colors and Graphics D) UI – Marketing, UX – Decoration
4. The "Define" stage in design thinking involves:
A) Creating color schemes B) Building user interfaces
C) Identifying and framing the user's problem D) Brainstorming wild ideas
5. Divergent thinking is best described as
A) Selecting one best solution B) Exploring many possible solutions
C) Narrowing focus on a single problem D) Eliminating irrelevant ideas
6. A key characteristic of divergent thinking is:
A) Judgment-based evaluation B) Structured analysis
C) Open-ended exploration D) Restricting alternatives
7. Divergent and Convergent thinking in design are:
A) Always separate processes B) Used together iteratively during design
C) Only relevant to artists D) Steps only in coding phases
8. Which rule is central to brainstorming?
A) Criticize all suggestions for improvement B) Avoid generating too many options
C) Withhold judgment on ideas during generation D) Focus only on realistic concepts
9. What does UX stand for in design terminology?
A) User Example B) User Experience
C) Universal Exchange D) Unique Experiment
10. Which of the following best defines UX design?
A) Creating fancy graphics B) Designing systems that provide meaningful user experiences
C) Optimizing code execution D) Building backend servers

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11. Which of these is NOT part of UX design methodology?
 - A) Research
 - B) Wireframing
 - C) Debugging
 - D) Testing
12. Which step comes after research in UX design?
 - A) Wireframing
 - B) User Needs Analysis
 - C) Debugging
 - D) High-fidelity Mockup
13. Which of the following is NOT a UX design activity?
 - A) Sketching
 - B) Prototyping
 - C) Marketing analysis
 - D) Usability testing
14. Why should UX designers consider business goals?
 - A) To replace user needs
 - B) To align product success with organizational success
 - C) To increase coding speed
 - D) To make testing unnecessary
15. An example of a user need is:
 - A) Easy navigation
 - B) Code optimization
 - C) Maximizing ROI
 - D) Cloud storage
16. Wireframing is best described as:
 - A) Finalized UI design
 - B) Low-fidelity design skeleton
 - C) Back-end architecture
 - D) Marketing strategy
17. What is the purpose of writing a problem statement in UX design?
 - A) To finalize the budget
 - B) To define the user's pain point clearly
 - C) To avoid ideation
 - D) To start coding immediately
18. Identifying appropriate research methods depends on:
 - A) Designer's preference
 - B) Research goals and resources
 - C) Business revenue
 - D) Code structure
19. Personas in UX design represent:
 - A) Marketing executives
 - B) Fictional but realistic user archetypes
 - C) Developers
 - D) Testers
20. Why are personas useful in UX design?
 - A) They replace real users
 - B) They guide design decisions based on user needs
 - C) They avoid usability testing
 - D) They finalize backend code

• PART - B: DESCRIPTIVE ANSWER QUESTIONS

Unit – I		Marks	BT*	CO*	PO*
1.	a) Define UI Design and UX Design. Explain how they differ in purpose, process, and deliverables with suitable examples.	08	L*2	1	1
	b) Analyze the stages of the Design Thinking process using a real-world product (e.g., a food delivery app). Discuss how divergent and convergent thinking are applied across these stages.	08	L3	1	1
2.	a) What is Observational Empathy? Illustrate its importance in understanding user needs and improving user experience	08	L2	1	1
	b) Conduct a comparative analysis of Brainstorming and Gamestorming methods. Design a short activity outline showing how these techniques can be used to generate ideas for an educational mobile app.	08	L3	2	1
3.	a) Apply the concept of branding to a real-world product or digital interface. Explain how consistent branding elements (such as logo, color palette, and typography) influence user perception and trust.	08	L3	2	1
	b) Explain the key requirements of a front-end style guide and discuss how each contributes to maintaining visual and functional consistency in user interfaces.	08	L2	2	1

Unit – II

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| 4. a) | Explain the concept of user experience (UX) and describe how the seven key factors collectively influence a user's interaction with a product. | 08 | L2 | 3 | 1 |
| b) | Explain the similarities and differences between Graphic Design and User Experience (UX) Design. Discuss how each discipline contributes to the overall visual appeal and usability of a product. | 08 | L2 | 3 | 1 |
| 5. a) | Explain the importance of understanding user needs and business goals in UX design. Discuss how aligning both ensures product success. | 08 | L2 | 3 | 1 |
| b) | Assume you are designing an online learning platform. Develop a simple user research plan that identifies tools, research methods, and user goals to guide the design process. | 08 | L3 | 4 | 1 |
| 6. a) | Explain the principles of wireframing and responsive design. Discuss how sketching and red routes contribute to creating user-friendly interfaces. | 08 | L2 | 4 | 1 |
| b) | Design and describe a low-fidelity prototype workflow for an e-commerce website. Explain how usability testing and iteration can improve its effectiveness. | 08 | L3 | 4 | 1 |

Unit – III

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| 7. a) | Explain what a problem statement is in the context of UX design. Describe the key features of a well-defined problem statement and discuss how a designer's Point of View (POV) helps in framing user-centered design challenges. | 08 | L2 | 5 | 1 |
| b) | Explain the role of flow diagrams and information architecture in structuring user interactions. How do they improve navigation and usability? | 08 | L2 | 5 | 1 |
| 8. a) | Describe the process of creating user personas. Explain how personas help in connecting research insights to design decisions. | 08 | L2 | 5 | 1 |
| b) | Develop a user story and scenario for an online food delivery platform. Explain how ideation techniques can be used to generate design solutions for identified user goals. | 08 | L3 | 5 | 1 |

BT* Bloom's Taxonomy, L* Level; CO* Course Outcome; PO* Program Outcome
